

Prashant Kumar

Vehicle dynamics, control and telemetry, proven trackside.

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Summary

Engineering graduate: **MSc Automotive Mechatronics** (Cranfield, 2026); B.Tech Electrical and Electronics Engineering, **UK 2:1 equivalent**. My thesis with **Zenvo Automotive** put **active aerodynamics** inside a minimum-lap-time optimal-control solver, and I was **Chief Engineer** of the team that won the **UTAC Challenge 2026**. **Three seasons trackside** (British GT Cup, BRSCC, Indian Formula 4) on the practical side of the car: setup changes, pit stops, tyres and data. Before Cranfield, 4.5 years of client-facing work presenting to stakeholders across time zones. I moved to the UK to work in motorsport.

Skills

Aerodynamics and performance: minimum-lap-time optimal control (CasADi / IPOPT), vehicle dynamics modelling, Pacejka tyre models, ride-height / downforce aero maps, IPG CarMaker, correlation of models against measured runs
Racing and data: Marelli WinTAX4, AiM RaceStudio3, MoTeC, ECUMaster, CAN bus, sensors and data loggers, telemetry and data analysis, tyre allocation
Software: MATLAB, Simulink, Simulink Coder, Python (NumPy, pandas), C, SQL / PostgreSQL, Git / GitHub, Linux
Electrical design and test: electrical subsystems, wiring harness, sensor integration, IoT product prototyping, test plans and validation gates

Education

Cranfield University, Cranfield, United Kingdom

Master of Science (MSc), Automotive Mechatronics

Sep 2025 - Sep 2026

- **Thesis with Zenvo Automotive**, *Implementable Active Aerodynamics in Minimum Lap Time Simulation*: extended a **minimum-lap-time optimal-control** framework (MATLAB, CasADi / IPOPT) with independent **front and rear wings** and four-wheel torque vectoring; 9 vehicle configurations solved at Barcelona-Catalunya and the Nürburgring.
- **Aerodynamics:** modelled the aero from the partner's **CFD** wing-angle sweeps and ride-height map, resolving the ride-height / downforce feedback loop into a smooth, differentiable map so **aero balance migrates per axle** with wing angle instead of one lumped coefficient; per-axle Pacejka MF5.2 tyres with combined slip.
- **Performance:** framed the result as **implementability, not headline gain**, measured how much of the free wing's optimum survives discrete positions, a mandated braking deployment and the actuator rate actually delivered, and showed a velocity-scheduled law designed against an assumed **60 deg/s** actuator **ceases to exist** at the rate actually available, under half that. The free wing recovers **0.24-0.43%** of a lap over the best swept static setting. Reported every delta against the method's measured numerical resolution; every change gated by **8 validation scripts and 26 unit tests**.
- Modules: Vehicle Dynamics, Advanced Control and Optimisation, Vehicle Control Applications (dSPACE HIL lab).

Manipal Institute of Technology, Manipal, India

Bachelor of Technology (B.Tech), Electrical and Electronics Engineering

Jul 2016 - Aug 2020

- CGPA 7.4/10 (**UK 2:1 equivalent**). **Team Manipal Racing** (Baja SAE): electrical subsystem design, wiring harness and sensor integration.

Work Experience

Tyre Performance Engineer (weekend contract)

Race Car Consultants, Suffolk, United Kingdom

May 2026 - Present

- **BRSCC Fiesta Junior and ST240** championships: own **tyre allocation** and lifecycle tracking for **12 customer cars** to the series technical regulations, each car signed off to the run plan before its session. **Zero tyre-reg penalties across 5 weekends**.
- Diagnose **CAN bus**, GPS and data-logger faults on **ECUMaster** systems, separating sensor or wiring faults from configuration faults before anything is replaced.

Junior Mechanic and Data (weekend contract)

SVG Motorsport, Gnosall, United Kingdom

Apr 2026 - Present

- **Ginetta G56 Evo, British GT Cup**: on race weekends, **setup changes** between sessions, **pit-stop operations**, and post-session data export and analysis fed back to the driver and coach.

Graduate Race Engineer

Indian Formula 4 Championship, Chennai, India

Aug 2025 - Sep 2025

- Re-engaged for a second season. Post-session **telemetry analysis** in **Marelli WinTAX4** alongside **AGS Formule 1** engineers to compare drivers and identify performance areas; relayed data and strategy to drivers in live sessions and supported setup changes against driver feedback.

- **Trained 30 incoming interns** on Mygale M21-F4 vehicle systems and pit-lane safety, using my telemetry platform as the teaching tool; on my **own initiative**, designed, tested and commissioned an **in-chassis trickle-charging unit** still in use by the team.

Analyst, Programmatic and Financial Operations

Publicis Global Delivery, Remote, India

Apr 2023 - Sep 2025

- Built **Python automation** for the daily data collection and pacing updates on the Anheuser-Busch InBev US account, cutting a daily manual process by **75%**; built and **presented** the dashboards the onshore US team used for spend decisions, working across time zones and raising data errors before they reached the client.

Product Development Manager

Purpuligo Technologies, Jamshedpur, India

Feb 2025 - Jul 2025

- Captured requirements from site visits and **stakeholder** interviews and wrote the **product specifications** (sensor integration, power management, connectivity) for IoT devices; ran **prototyping** from concept to **field-tested units** in mining and agricultural environments, iterating on field data.

Graduate Junior Mechanic

Indian Formula 4 Championship, Chennai, India

Aug 2024 - Nov 2024

- Selected by written examination. Prepared Mygale M21-F4 cars for every session with **MP Motorsport** and **Evans GP** engineers; diagnosed and replaced **on-car electronics** under failure (temperature and pressure sensors, throttle-position sensor, data loggers, a dash after calibration loss).

Senior Business Analyst, Advanced Analytics

Merkle Sokrati, Pune, India

Jul 2022 - Feb 2023

- Wrote **SQL** on Google Ads Data Hub to build multi-channel attribution for four clients, turning each client's business question into a hypothesis, a query and a dashboard; **promoted** into this team from the role below.

Business Analyst, Programmatic

Merkle Sokrati, Pune, India

Nov 2020 - Jun 2022

- Post-sale technical contact for a Singapore telecoms client: ran **status meetings** with internal and external **stakeholders**, resolved discrepancies between 1st- and 3rd-party **reporting** systems, and delivered post-campaign analysis within a two-business-day deadline.

Projects

UTAC Challenge 2026 (Kia Niro EV): IPG CarMaker, Simulink, Python, CAN

2026

- **Overall winners. Chief Engineer:** led the **9-person** Cranfield team (weekly cadence, test-facility and dynamometer access, the link to supervisors), built and ran the test plan (static, dynamic and track characterisation) and wrote the **Python CAN-bus logger** that captured every run.
- Built the **IPG CarMaker** vehicle model from the team's own measurements and **correlated it against the measured** cornering and slalom runs; with no tyre data or wheel-speed sensor on the car, built the friction estimator's lookup table from **50+ CarMaker surface-friction runs**; ran the team's shared GitHub repository.

Active-aero controller and 6-DOF vehicle plant: Simulink, Simulink Coder, C

2026

- Turned the thesis wing law into a **100 Hz** discrete controller against a **6-DOF vehicle plant**; generated C proved **bit-equivalent** to its model under a **9-gate acceptance suite** that caught two defects review had missed and showed a library tyre block structurally could not represent the car.

indianf4championship.com (telemetry platform): Python, Flask, PostgreSQL, Plotly

2024 - Present

- Built on my **own initiative**, and still run alone, the platform Indian Formula 4 race engineers use at the circuit: ingests Marelli WinTAX4 and AiM RaceStudio3 exports and overlays up to **4 drivers on a common distance basis** so **lap-time delta** is attributable corner by corner; Docker on Azure; the championship's **intern training tool**.

FullModelSim (vehicle simulation app): Python, CasADi / IPOPT, PySide6

2026

- Ported a **23-state vehicle model** and optimal-control solver from MATLAB to **object-oriented Python** (5,600 lines) behind a PySide6 desktop application, solver run out of process; root-caused a convergence stall to a missing scaling option.

Additional Information

Leadership and communication: Vice President, Motorsports Society of Cranfield (race screenings, karting, track trips, Q&As with team engineers); Student Ambassador, Cranfield University; life-skills trainer for 56 students (Jharkhand Health, Educational and Charitable Trust).

Right to work: Graduate visa valid to Jan 2029; eligible for Skilled Worker sponsorship.

Full UK driving licence; available for race-weekend travel and open to relocation. Languages: English (C2), Hindi (native).